Nikki Taylor

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Senior full-stack software engineer with a track record of independently building high-impact internal tools and research platforms for ML engineers and games user researchers.

EXPERIENCE

Senior Technical Operations Developer, PlayStation Studios — February 2024 – Present

- Sole owner and engineer of a full-stack internal research operations and data analysis platform used by 70+ user researchers and analysts. Refactored a legacy PHP/jQuery application into a modular MVC architecture with structured APIs, OpenAPI documentation, redesigned Bootstrap-based interfaces, and newly introduced React frontend capabilities.
- Rapidly designed and shipped new tools for manipulating research data, including an AI text analysis tool that automated 900+ qualitative research reports in a year, saving researchers 2+ working days per report.
- Developed a real time survey data ingestion pipeline and UI tool enabling real-time analysis during user tests and integration with analytics tools across the organization. Deployed a self hosted Prefect instance with a real-time Python ETL pipeline in addition to a PHP API and UI for configuring event workflows.
- Partner directly with researchers to identify high-impact opportunities for workflow automation, rapidly prototype AI-assisted solutions, and iteratively refine tools based on user feedback.
- Independently created product planning and engineering standards in an unstructured environment, from defining the high level vision and roadmap for the tools team to implementing sprint planning, version control, automated deployment, linting, alerting, and AI development procedures.

Machine Learning Engineer, Headspace — November 2021 – February 2024

- Owned ML platform and developer operations workflows for the team, managing AWS infrastructure and building command line utilities that streamlined SageMaker model scaffolding, deployment, and experimentation workflows for ML engineers.
- Built reusable CI abstractions enabling rapid implementation of automated unit and integration testing across 10+ production microservices.
- Created reusable infrastructure modules using TypeScript and AWS CDK to simplify provisioning and orchestration of batch ML workflows, improving developer velocity.
- Delivered an LLM-based recommendation system using clinical free-text data, leading the project end-to-end from exploratory analysis and model development through production deployment. Participated in responsible AI and data governance reviews to ensure safe handling of sensitive clinical data and decision-making systems.

Graduate Machine Learning Researcher, Stanford University Helix Group — June 2020 – June 2021

- Developed an embedding based machine learning method to classify genes associated with the pathogenesis of nonalcoholic steatohepatitis used to validate results from in vivo experiments. Publication: *Mapping transcriptional heterogeneity and metabolic networks in fatty livers at single-cell resolution, iScience, 20 January 2023.*
- Led communications of complex technical and analytic results to broad groups of stakeholders, including funders, corporate partners, and academic researchers.

EDUCATION

Stanford University

M.S. Biomedical Data Science — 2020 – 2021

B.S. Human Biology - Computation and Neuroscience — 2016 – 2020

Awards: Phi Beta Kappa, Joshua Lederberg Award for Academic Excellence **GPA**: 4.03

Graduate Capstone: *Deep Learning Assisted Parameter Mapping for the Assessment of Stroke Lesions* - Segmenting stroke location in brain images (DWI and MRP) using UNET and Convolutional Neural Networks with Keras and Tensorflow.

SKILLS

Python, PHP, Javascript, Typescript, HTML/CSS, React, AWS, half marathons, handstands